INCH-POUND
MIL-PRF-3098/104F
6 March 2013
SUPERSEDING
MIL-PRF-3098/104E
16 January 2006

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR102/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 50 MHz to 125 MHz; fifth overtone; noncontrolled; series resonance.

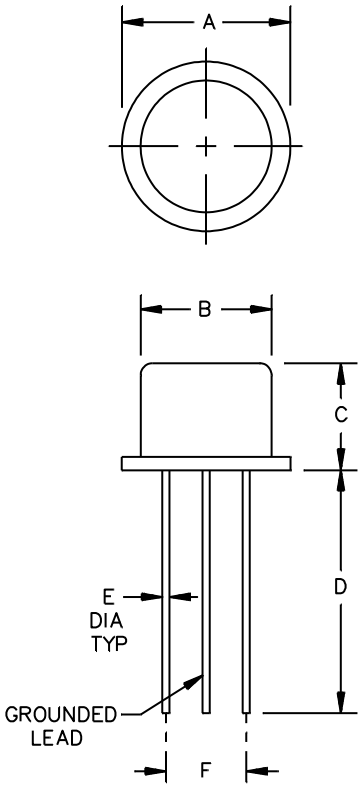


FIGURE 1. Crystal unit - CR102/U.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:51Z
Check the source to verify that this is the current version before use.

Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.420	---	10.67
B	---	.326	---	8.28
C	---	.265	---	6.73
D	.600	---	15.24	---
E	.016	.020	.406	.508
F	.192	.208	4.88	5.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR102/U, crystal unit - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:51Z
Check the source to verify that this is the current version before use.

REQUIREMENT:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50 MHz to 125 MHz, inclusive.

Frequency tolerance: ±25 ppm.

Equivalent resistance: 50 MHz to 90 MHz: 50 ohms, maximum.
90 MHz to 125 MHz: 60 ohms, maximum.

Mode of oscillation: Fifth overtone.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 7 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Vibration: Method 204, MIL-STD-202, test condition B.

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

| The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:51Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 84, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2012-015)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:51Z
Check the source to verify that this is the current version before use.